IoT Cat Feeder

An automatic, timed cat feeder with a web dashboard on the ESP8266

Project Proposal

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November 2026

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Chapter 1 Introduction

1.1 Motivation

A cat that is fed by hand eats at the convenience of the owner, not at a fixed time. When the owner leaves for work in the morning and returns in the evening, the usual workaround is to pour a large portion into the bowl and let the cat graze all day. Two problems follow. The food sits in the open bowl for ten hours or more, and the cat eats the entire portion before noon and then begs for the rest of the day.

The second problem is portion control. Dry food is calorie dense, and a cat that free feeds commonly gains weight. Veterinary guidance for an adult cat of average weight is a measured daily ration split across several meals, not an open bowl. An owner who is not at home at lunchtime cannot deliver the middle meal.

Commercial automatic feeders solve the timing problem, but most of them depend on a cloud service and a mobile app to program the schedule. If the vendor changes the app, or the household Wi-Fi is replaced, the device becomes a plastic box. For a student project the interesting engineering question is whether the same behaviour can be produced on a microcontroller that the owner can read, repair, and reflash.

1.2 Purpose

This project builds a cat feeder that dispenses dry food at three fixed times per day and also on demand from a phone. The device serves its own web dashboard over the local Wi-Fi network, so no account, no cloud service, and no vendor app are required. The owner opens a browser, sees the current state of the flap and the number of meals served today, and can trigger a feeding with one tap.

The project is also a study of the boundary between software and hardware. The interesting part is not the web page. It is the electrical and timing behaviour that the web page depends on: a servo that stalls and browns out the board, a clock that is wrong until the network answers, and a flap that must never stay open.

1.3 Target Users and Intended Experience

Two users with different needs must both be served without one damaging the other.

1.3.1 Persona: The owner

Table 1. Owner persona

Attribute	Detail
Name	Nok
Age	29
Occupation	Office worker, Chiang Mai
Background	Lives in a rented room with one adult cat. Leaves at 07:30 and returns at 18:30. Uses a phone for everything. Has no tools beyond a screwdriver.
Goals	Feed the cat on a predictable schedule. Check from work that the morning meal was served. Add an extra meal when the cat begs.
Pain points	Cannot come home at lunch. Forgets whether the morning meal was served. Does not want another app account, and does not want to learn a configuration language.
Experience requirement	Open one page, read five values, tap one button. No login, no setup wizard, no pairing step.

1.3.2 Persona: The cat

Table 2. Cat persona and constraints

Attribute	Detail
Subject	One adult domestic cat, 4 kg
Behaviour	Eats at the sound of the flap. Approaches the device when it is near the bowl.
Constraint	The cat must not be able to open the flap with a paw, and must not be trapped by it.
Experience requirement	A meal arrives without the cat having to touch anything. The flap cycle is short enough not to startle the cat away from the bowl.

1.4 Project Scope

The scope is set so that the work is complete and demonstrable on 4 November 2026. Features that cannot be finished and tested by 2 November are moved to Future Work rather than shipped half-built.

Table 3. Scope of the delivered system

In scope	Out of scope
Scheduled feeding at three times per day using NTP time	Cloud storage and a hosted dashboard
On-demand feeding from a browser on the same network	A native mobile application
Flap status, daily meal count, last meal time and source	User accounts and authentication
Automatic flap close after a fixed open interval	Wet food dispensing
Cat presence detection with an ultrasonic sensor (target)	Camera or two-way audio
Hopper level from a downward ultrasonic sensor	Portion control by weight with a load cell
	Support for more than one feeder or more than one cat

Portion control left the scope on 2 November, when the header pins of the amplifier were found to be unsoldered. It moved to Future Work rather than ship half-built. Presence detection is wired and integrated, and it ships if it passes the test on 3 November. A requirement that is written and tested earns credit, while a half-connected sensor does not.

1.5 Objectives

1. Deliver food at 08:00, 13:00, and 18:00 local time, using time obtained from a network time server rather than a countdown timer.
2. Serve a dashboard from the ESP8266 itself, reachable from a phone on the same Wi-Fi network in under two seconds.
3. Close the flap automatically after every feeding, in all code paths, including the path where the browser disconnects immediately after sending the command.
4. Record the time, count, and trigger source of each meal so the owner can confirm a feeding happened.
5. Detect a cat within 60 cm and dispense a meal in under two seconds, at most once per 30 minutes.
6. Document the design so another student can rebuild the device from the repository alone.

Chapter 2 Review and Analysis

2.1 Tool and Technology Review

Each decision below was made against at least one alternative. The rejected option is recorded so the reader can see the trade that was accepted.

Table 4. Technology decisions

Decision	Chosen	Rejected	Rationale
Microcontroller	NodeMCU v3, ESP8266	Arduino Uno with ESP-01, ESP32	Already owned. Has Wi-Fi and a 3.3 V regulator on board. The ESP32 is better but adds cost and no requirement needs it.
Actuator	Servo SG90	Stepper motor, solenoid	The servo holds an angle without power cycling and needs one PWM pin. A solenoid slams and needs a driver transistor.
Time source	SNTP over Wi-Fi	DS3231 real-time clock	No extra part, no battery, and the time is correct after every boot. An RTC drifts and needs a battery holder.
Transport	HTTP on the device	MQTT broker, Blynk	HTTP needs no broker and no account. Blynk hides the implementation behind a vendor widget, which removes the work the assessment is looking for.
Dashboard	Embedded HTML and JavaScript in PROGMEM	SPIFFS files, external CDN	One flash image, no filesystem, no internet dependency, and the page cannot lose its assets.
Presence sensing	HC-SR04 ultrasonic	FC-51 infrared proximity	Ultrasonic reports distance, which supports both presence and level. The FC-51 is binary and is fooled by a cardboard hopper wall.
Portion control	Open-loop flap timing	HX711 with 1 kg load cell	The load cell measures the actual mass, but the amplifier header pins were not soldered before the deadline. Open-loop timing ships, and the closed loop moves to Future Work.

2.2 Risk Analysis

Table 5. Risk register

ID	Risk	Likelihood	Impact	Mitigation
R-01	Servo stall browns out the board and reboots it mid-meal	High	High	Power the servo from the powerbank rail, not from 3V3. Add a 470 uF capacitor across VIN and GND. Test the worst case with servo_test.ino before integrating.
R-02	Flap stays open and the hopper empties	Medium	High	The close is driven from millis() in the main loop, not from a network callback, so a dropped client cannot strand the flap.
R-03	Clock is wrong or jumps backward after a resync	Medium	Medium	Ignore the schedule until the clock passes a fixed epoch. A duplicate meal inside the same minute is prevented by a minute key.
R-04	Ultrasonic readings oscillate on a kibble surface	High	Medium	Sample ten times per measurement and use the median. Reject echoes outside the hopper depth.
R-05	Wi-Fi in the examination room blocks the device	Low	High	The dashboard is reachable in the local network, so the demonstration does not depend on internet access. A phone hotspot is the fallback.

2.3 Estimated Cost

Prices are estimates from general retail listings and are not yet confirmed with the supplier.

Table 6. Estimated bill of materials

Item	Quantity	Unit (baht)	Total (baht)
NodeMCU v3 (already owned)	1	0	0
Servo SG90 (already owned)	1	0	0
Capacitor 470 uF 16 V	1	10	10
HC-SR04P ultrasonic sensor	2	80	160
HX711 amplifier with 1 kg load cell	1	160	160
Breadboard, jumpers, resistor (already owned)	1 set	0	0
Cardboard, tape, food-safe bowl	1 set	50	50
Powerbank 5 V 2 A (already owned)	1	0	0
Total			**380**